



Seminar on User-Centered Programming Interfaces

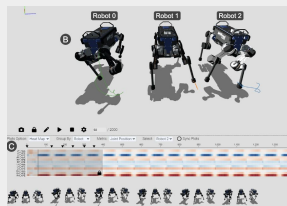
Course Overview

Prof. April Wang

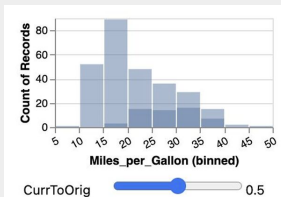
2026.09.17

The PEACH Lab at ETH Zürich designs interactive systems to **make programming environments more usable** both in professional and educational contexts.

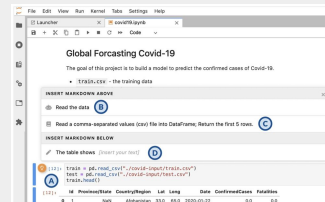
Professional Context



robotics training

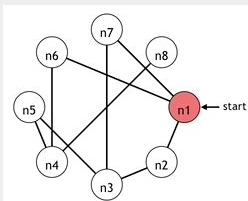


data analytics

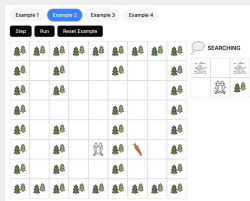


computational notebooks

Educational Context



algorithm visualization



visual programming



augmented reality



<https://peachlab.inf.ethz.ch/>



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Introduction

- Who are you?
- Why are you interested in this topic?

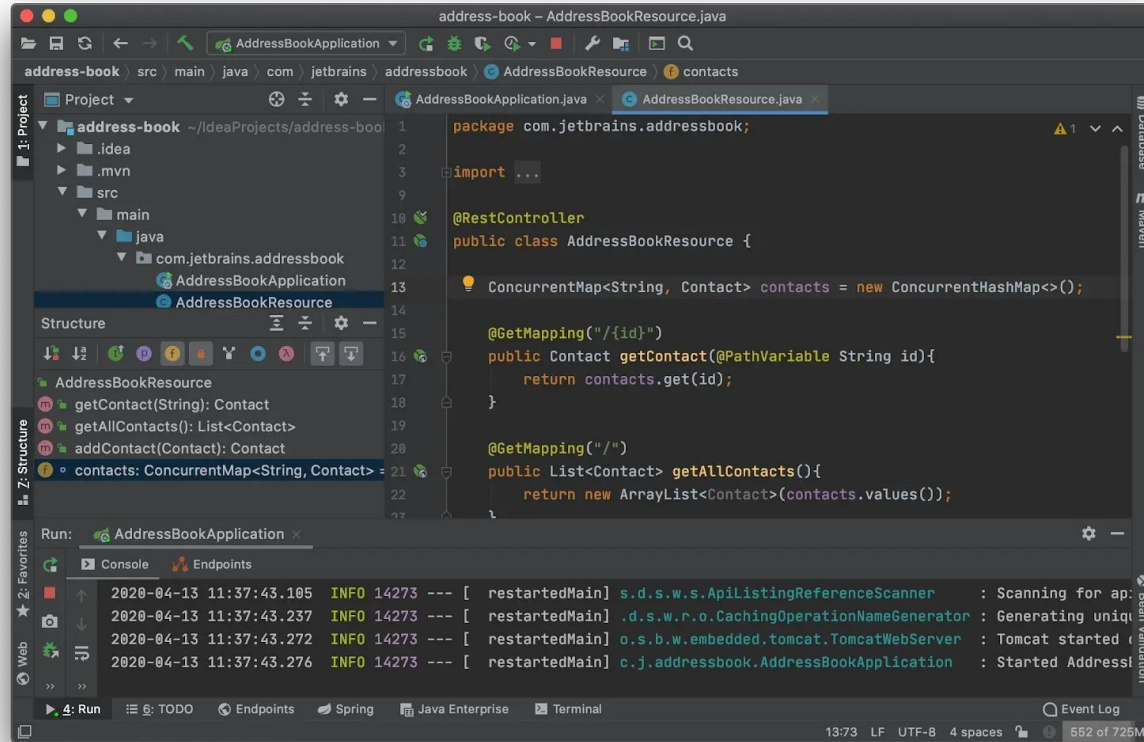
Course Overview

Seminar on User-Centered Programming Interfaces ([263-4902-00L](#))

Time	Thu, 16:15 - 18:00
Location	CAB G52
Language	English
Office Hours	Doodle Link
Seminar Website	https://peachlab.inf.ethz.ch/teaching/ucpi2026/

How should we design interfaces that help people understand, create, and modify programs?

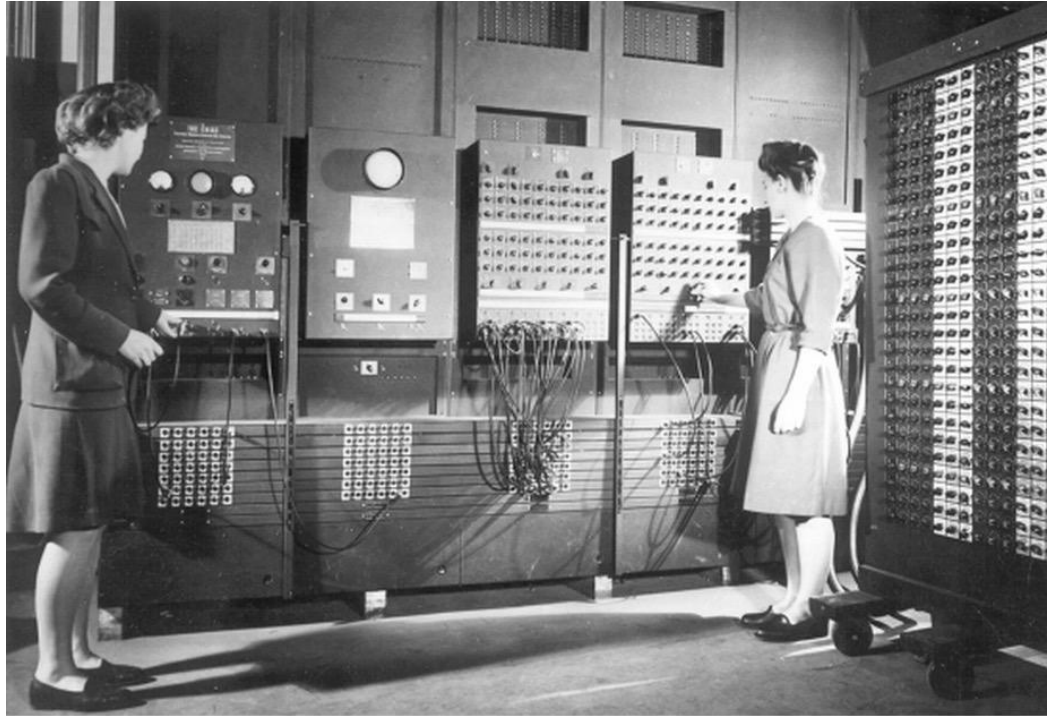
What is a programming interface?



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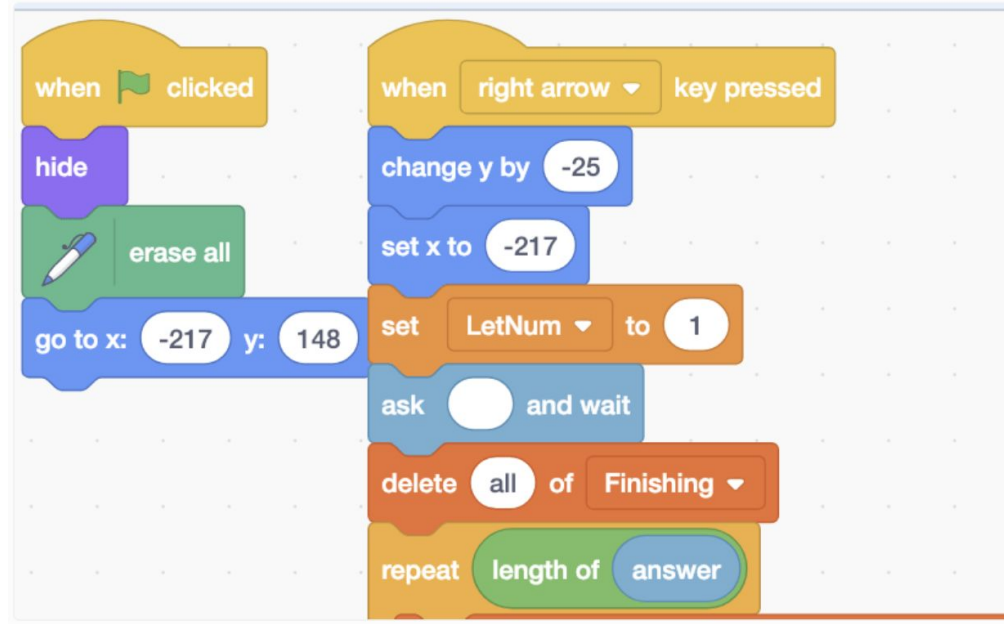


What is a programming interface?

	A	B	C	D	E	F	G
1	expense	amount					
2	Textbook	€ 30					
3	Highlighter	€ 4					
4	Coffee	€ 2					
5							
6	total	=SUM(B2:B5)					
7							

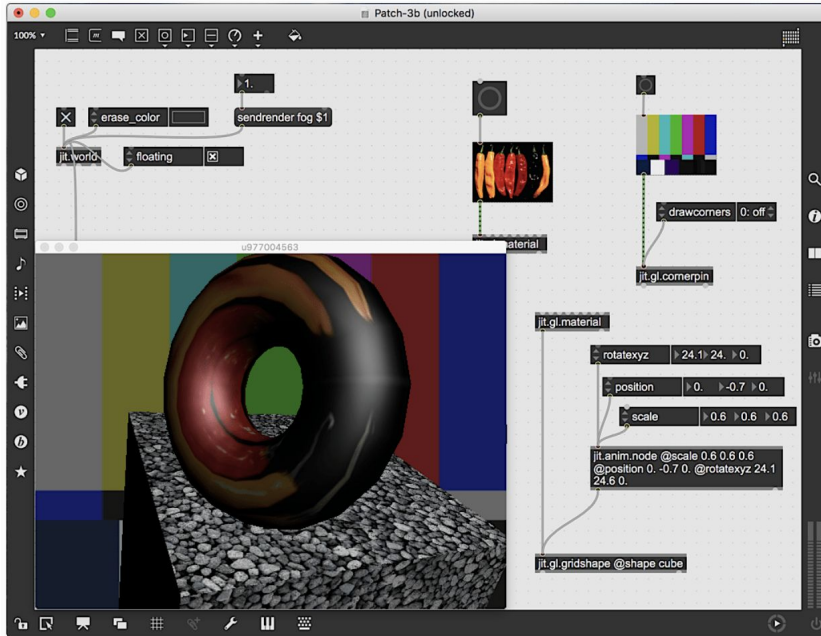
getting insights from data

What is a programming interface?

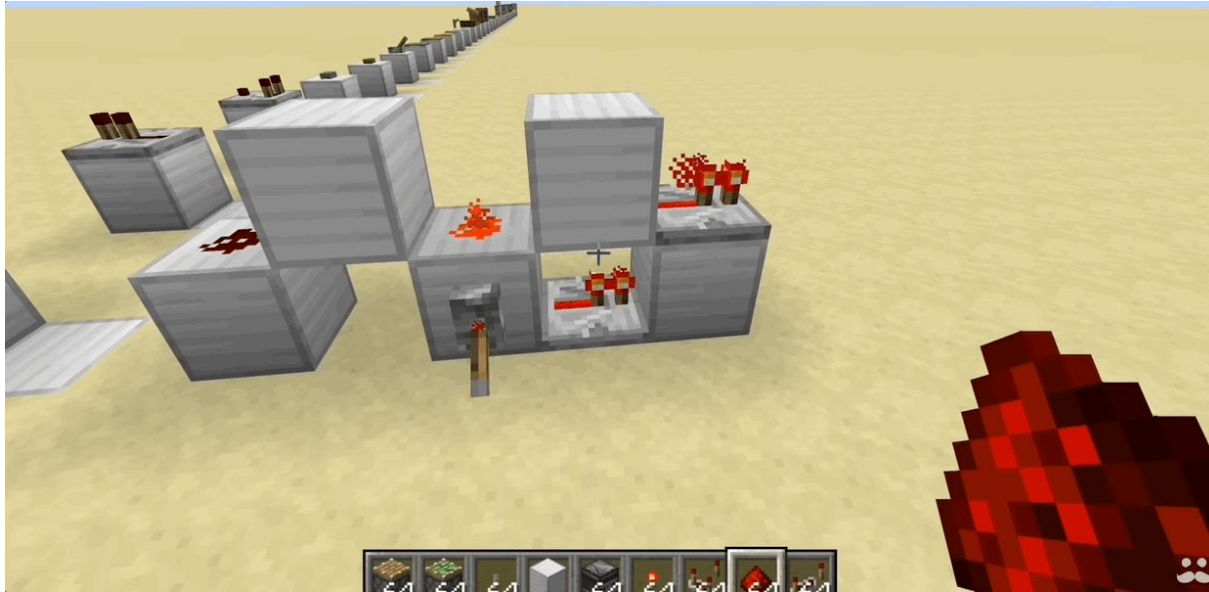


developing problem-solving in young minds

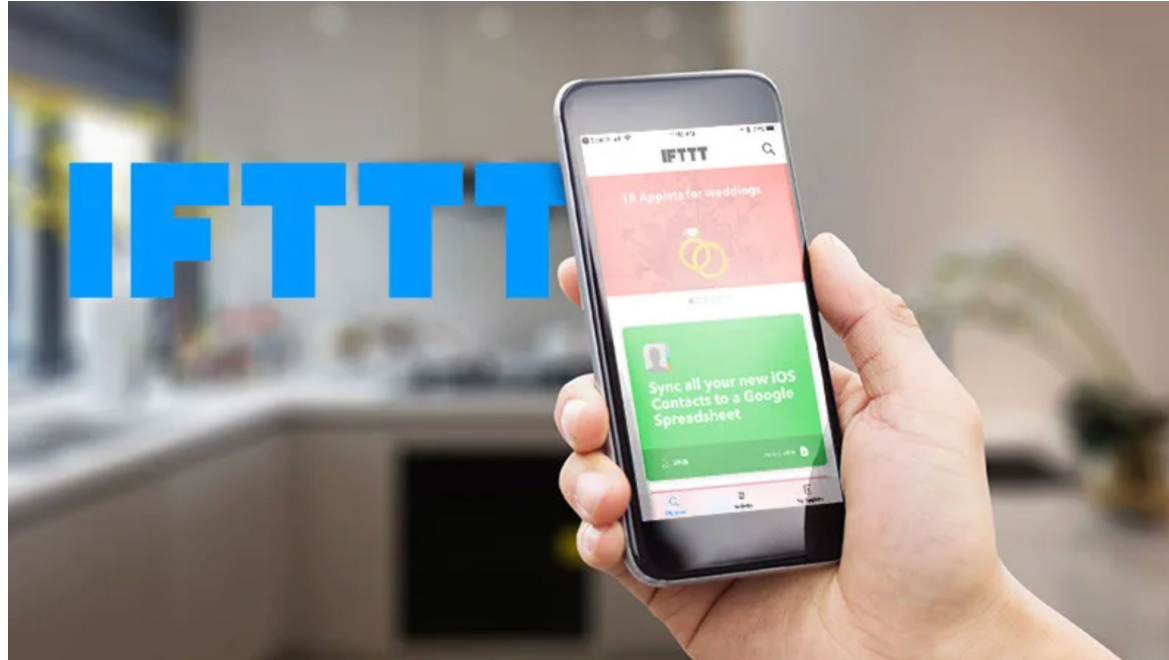
What is a programming interface?



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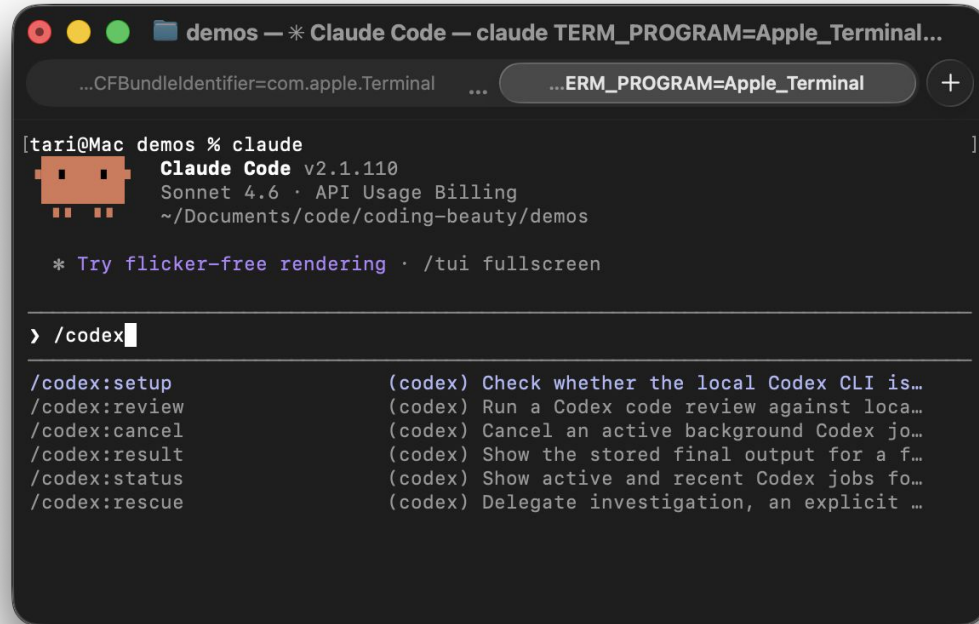


What is a programming interface?



IFTTT: If this, then that

What is a programming interface?



The screenshot shows a macOS terminal window titled "demos — * Claude Code — claude TERM_PROGRAM=Apple_Terminal...". The terminal displays the output of the `claude` command, which includes the Claude Code logo, version information (v2.1.110), and the path to the current directory. Below this, a list of commands and their descriptions is shown, such as `/codex:setup` for checking the local CLI, `/codex:review` for running a code review, and `/codex:rescue` for delegating investigation.

```
[tari@Mac demos % claude]
Claude Code v2.1.110
Sonnet 4.6 · API Usage Billing
~/Documents/code/coding-beauty/demos

* Try flicker-free rendering · /tui fullscreen

> /codex

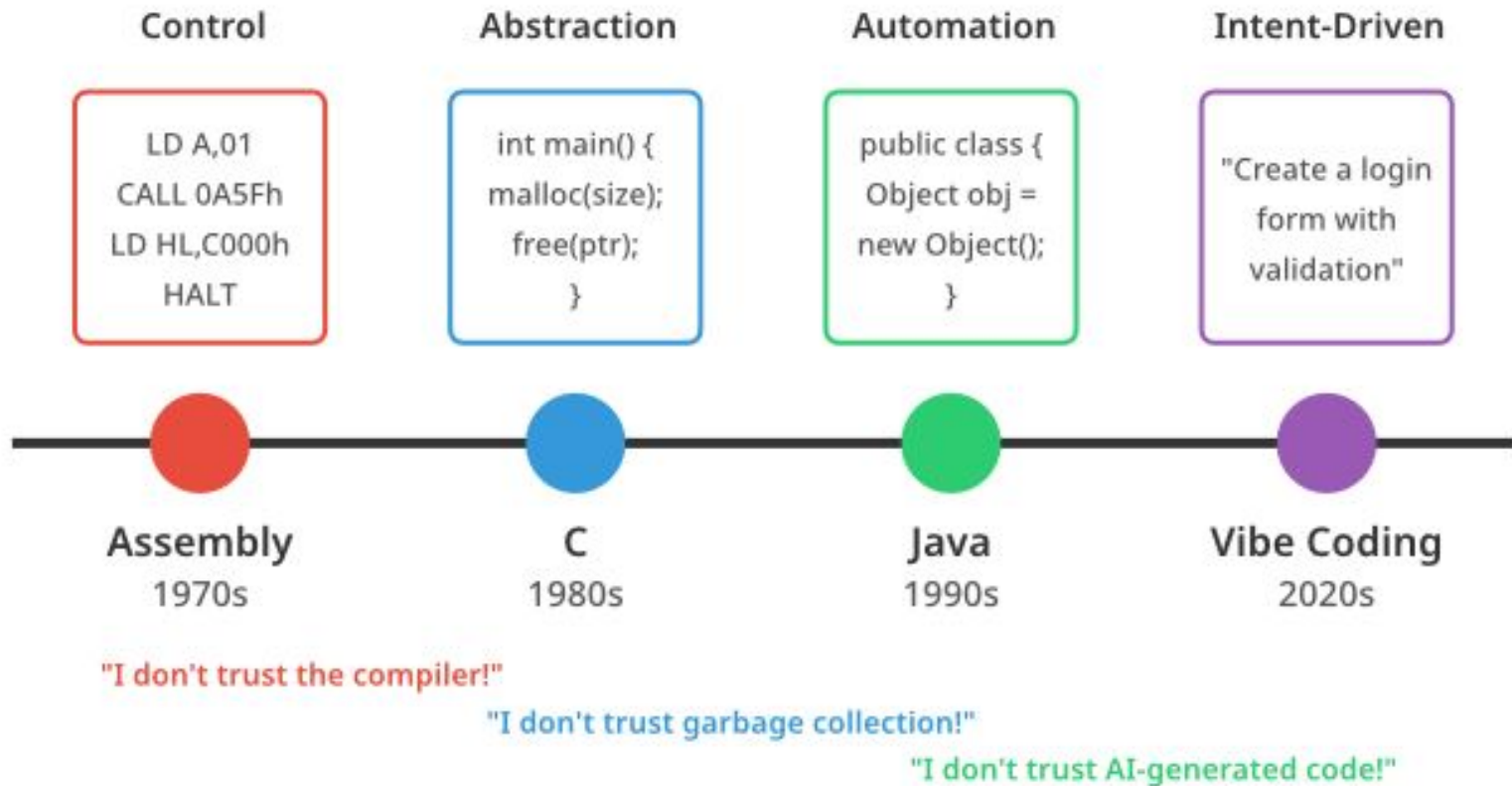
/codex:setup          (codex) Check whether the local Codex CLI is...
/codex:review         (codex) Run a Codex code review against loca...
/codex:cancel         (codex) Cancel an active background Codex jo...
/codex:result         (codex) Show the stored final output for a f...
/codex:status         (codex) Show active and recent Codex jobs fo...
/codex:rescue         (codex) Delegate investigation, an explicit ...
```

The lens for this year

Pre-LLM vs. LLM-based

Questions we will keep asking

- How do users express intent?
- How do users understand system behavior?
- Who has control?
- What should the system infer?



Topics to Cover

- Program Understanding
- Code Editing
- End-User Programming
- Debugging
- Program Histories / Versions
- Collaborative Programming
- Interactive Graphics & Articles
- Teaching Programming

W	D	Topic	Presenters
1	9.17	Course Logistics	
2	9.24	Topic Overview: Redefining Programming	
3	10.1	Break	
4	10.8	Program Understanding	
5	10.15	Code Editing	
6	10.22	End-User Programming	
7	10.29	Programming by Demonstration	
8	11.5	Guest Lecture	
9	11.12	Debugging	
10	11.19	Managing Versions and Alternatives	
11	11.26	Collaborative Programming	

How each week works

- 2 presentations on the same topic, one pre-LLM, one LLM-based
- Presenter
 - ~10 min — Background + related work
 - ~15 min — **Revibed** prototype demo
 - ~5 min — Original evaluation
 - ~15 min — Discussion

Everyone reads both papers before class.

What is Revibe?

- Revibe = reimplement an interactive research artifact from its paper using contemporary AI-assisted programming tools

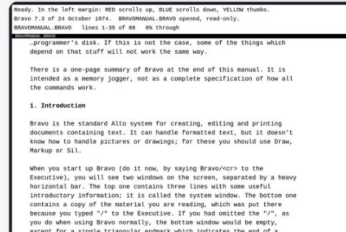


<https://revibe-hci.github.io/>

Eight scroll bars, 1974–1982

Re-created from SIGGRAPH Video Review, Special Issue #57 — “All the Widgets” (Brad A. Myers, Carnegie Mellon University, April 1990)

“The widget that allows you to control what part of the information is in the window is called a scroll bar, since you can think of the information as being on a large scroll.”



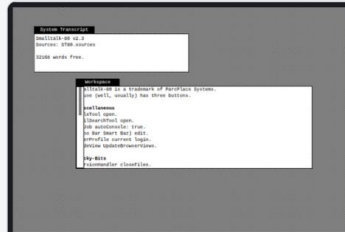
01 • 1974

Bravo

Xerox PARC

An invisible scroll bar. Only the cursor tells you it is there.

tape 03:07



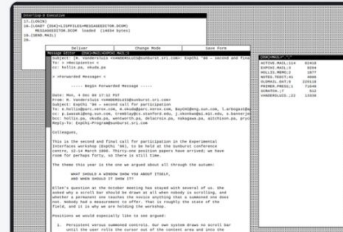
02 • 1977

Smalltalk

Xerox PARC / ParcPlace Systems

Pops up on hover; three vertical columns; the first proportional thumb.

tape 03:58



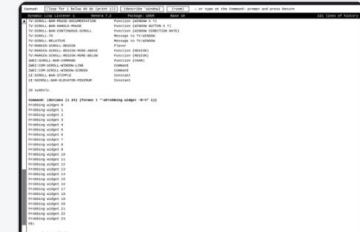
03 • 1980

Interlisp-D

Xerox Corporation

Appears only when you ask, by pushing into the margin.

tape 04:27



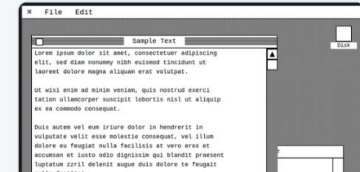
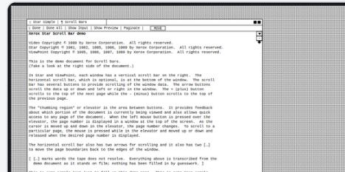
04 • 1980

Dynamic Windows

Symbolics, Inc.

The document grows under your hand, and the screen says what the buttons do.

tape 05:21

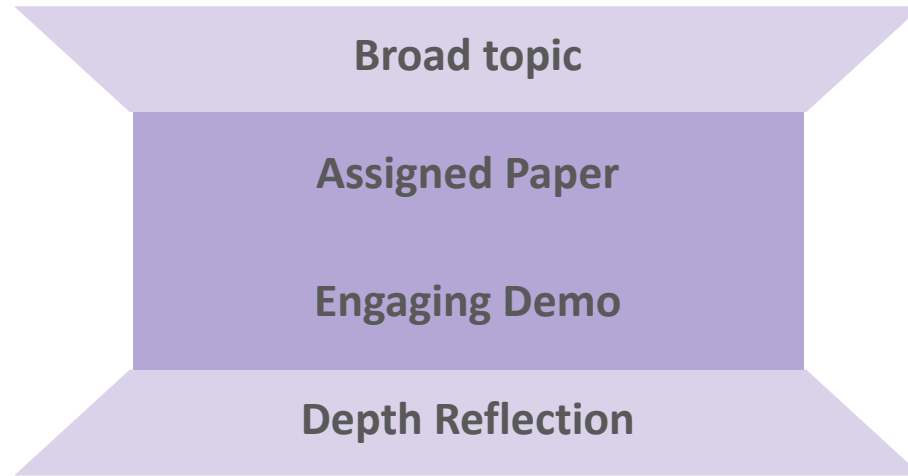


<https://scrollbardemo.netlify.app/index.html>

What makes a good Revibe?

- **Focus on the core interaction**
 - ✓ What the user can do
 - ✓ How the system responds
 - ✓ What makes the interaction distinctive
- **Not required:**
 - ✗ Full production system
 - ✗ Exact implementation
 - ✗ Pixel-perfect recreation

What makes a good paper presentation?



Final Submissions: Interactive blog post

Turn your Revibe into an interactive explanation of the research topic

- Expand your reading to ≥ 5 substantive sources
- Interactive article to explain the topic
- Help readers experience an idea
- Descriptive enough to be self-contained

Grading + Expectations

- **Presentation: 50%**
- **Participation: 20%**
 - Sign up sheet
 - Every presenter can nominate 5 MVP participants
- **Interactive blog post: 30%**

Reading List

- A detailed introduction of all papers:
https://docs.google.com/presentation/d/1Ypxq1TMqAC3N8wQkoQTYtKR3iOVnqKHVR8C_DYOtuJY/edit?usp=sharing
- Sign up sheet: <https://forms.gle/5uiifeAKAx89GVXZ6> (Please fill it in by September 24, the end of the day)